LAW ENFORCEMENT IT & DIGITAL FORENSICS GUIDE

SHA-256 CRYPTOGRAPHIC HASH DECODING & EVIDENCE VERIFICATION PROTOCOL

I. OVERVIEW OF SHA-256 CRYPTOGRAPHIC EVIDENCE INTEGRITY

In digital forensics and criminal prosecution, maintaining an unalterable chain of custody requires cryptographic hashing. A SHA-256 digest is a 256-bit (64-character hexadecimal) cryptographic witness string generated via FIPS PUB 180-4.

Any modification to the underlying coordinate dataset—even changing a single digit or comma—completely alters the resulting hash value (the 'avalanche effect'), providing absolute proof of data integrity.

II. LAW ENFORCEMENT IT DECODING & VERIFICATION PROTOCOL

Police IT departments and digital forensics units can independently verify any Project REVVEL dataset using standard command-line tools or Python scripts:

Target File Name	File Description	Linux / macOS Verification Command	Expected SHA-256 Witness Hash
master_spatial_overlay_and_pcr_prosecution_v51.csv	Master Prosecution Matrix	sha256sum master_spatial_overlay_and_pcr_prosecution_v51.csv	07c9863060868b7b75eb8e6b4b859bed...
master_sar_targets_v53.geojson	GeoJSON Web Spatial Layer	sha256sum master_sar_targets_v53.geojson	30df542c-2731-420b-9137-17b63479...
sunpeaks_master_sar_database.db	Master SQLite Database	sha256sum sunpeaks_master_sar_database.db	5cb1f2c7-c61c-4b4b-b0e9-8c794b70...

III. STEP-BY-STEP IT AUDIT INSTRUCTIONS

1. Open Terminal or Command Prompt on the evidence analysis workstation. 2. Navigate to the directory containing the downloaded Studio artifact. 3. Run 'sha256sum ' (Linux) or 'shasum -a 256 ' (macOS/PowerShell). 4. Compare the output hash string against the witness hash published in the court filing.